

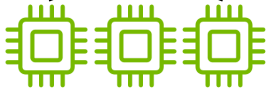
Stage 1: Task Pool

Class 1: Single kernel

Class 2: Module

Class 3: Application

Class 4: Unsolved
Challenge



Ampere Hopper Blackwell

Task Spec
HW metadata

Stage 2: LLM Generation



One-shot

Prompt

CUDA code



Tool-Augmented



Profiler



Logs



HW specs



Iterative Refinement



Compile



Run & Profile

Generated
CUDA code

Stage 3: Multi-Stage Eval



①

Compilation



Pass



Fail



②

Correctness

Torch.allclose()



Pass



Fail



③

Semantic Validity



Pass



Fail



④

Performance



Speedup



⑤

Arch Performance



Arch-specific
Metrics

Diagnostic Report

- capability gaps
- Per-domain breakdown
- Failure mode analysis